Project Overview

Platform: RentMate — AI-powered roommate matching + affordable housing for students and international newcomers in Hong Kong
Track: 3. Smart Living, Culture & Belonging
SDG: SDG 11 — Sustainable Cities and Communities
Social Issue: Equitable access to housing information and roommate matching for youth and international newcomers through a trusted digital community

Problem Statement

How might we improve the safety, convenience, and effectiveness of finding compatible roommates and affordable rental housing for university students and international newcomers in Hong Kong?
Context

- Hong Kong is experiencing an influx of international students and technopreneurs, especially with the Northern Metropolis 5-year plan bringing new housing and population to the border areas
- Existing platforms (28Hse, Nestia, Facebook groups) solve housing OR roommates separately, never both together
- Exchange and new students arrive blind — they can't visit flats, can't verify listings, can't assess roommate compatibility
- The rental market is opaque, agent-dominated, and expensive — students pay $5,000-10,000 in agent fees for doing almost nothing

End Goal
Fast onboarding → find a roommate and a house in days, not weeks

Problem Breakdown

#	Problem	Who It Affects	Current Solution	Gap
1	Difficulty finding available housing	Exchange students, new students	28Hse, Nestia, Facebook groups	Can't verify remotely, no trust signals
2	Finding compatible roommates	All students needing to share rent	Facebook groups, word of mouth	Random matching, no compatibility data
3	High agency fees	All renters	Pay 1 month rent as fee ($5,000-10,000)	Agents add little value for students
4	Lack of transparency	All renters	None	No way to verify size, condition, fairness of price
5	Safety and lifestyle mismatch	Everyone	None	No verification, no compatibility screening
6	Information inequality	International students	None	Local students have networks, newcomers have nothing

1. Target Users

Primary Users: non local students

User	Description	Key Need
Exchange students	Arriving from overseas, 1-2 semesters, don't know HK	Remote flat assessment, quick roommate match, trust
New local university students	First-year, moving from home, limited budget	Affordable housing, compatible roommates
International postgrad students	Staying 1-2 years, higher budget but still need value	Quality housing, professional roommate match
Young technopreneurs	New to HK via Northern Metropolis, need housing fast	Quick onboarding, flexible terms

Secondary Users

User	Description	Key Need
Existing tenants with a spare room	Want a compatible roommate, not just any tenant	Compatibility matching, verified profiles
Landlords	Want reliable tenants, minimal hassle	Verified tenants, less vacancy time
Student housing offices	Universities managing exchange housing	Better tools for incoming students

2. Research: Data & Pain Points

Market Data

- HKU alone hosts ~5,000 exchange students per year
- Average student rent: $4,500-7,000/month for a shared room
- Agent fee: 1 month rent ($5,000-10,000) — often unavoidable
- 70% of exchange students report housing as their biggest stressor (anecdotal, but consistent across forums)
- Northern Metropolis plan will bring 2.5 million people to new development areas by 2030

Pain Point	Evidence
"I signed a lease sight-unseen and the room was half the size advertised"	r/HongKong, r/HKU frequent posts
"My roommate parties every night and I have exams"	Facebook groups, student forums
"I paid $8,000 agent fee and the agent disappeared after"	Consumer complaints, news articles
"I can't tell if a listing is a scam from photos alone"	Common fear among exchange students
"Local students have friend networks to find roommates, I have nothing"	Exchange student testimonials
"I found a flat but need a roommate to afford it — where do I even look?"	Most common post in housing groups

Existing Platforms vs. Gap

Platform	What It Does	What It Doesn't
28Hse	Property listings	No roommate matching, no verification, no AI
Nestia	Property listings + virtual tours	No roommate matching, agent-dependent
Facebook Groups	Roommate/flat posts	No matching algorithm, no trust, chaotic
Squarefoot	Premium listings	Not student-focused, expensive
Flatmate.com	Roommate matching	Not HK-specific, no housing listings

The Gap: No platform combines housing listings + roommate matching + AI-powered trust + student-focused pricing in one place.

3. Potential Solutions ×3

~~Solution A: "MatchFirst" — Roommate-First Approach~~
~~Core idea: Find your roommate first, then find a flat together.~~
~~Target: roommate compatibility~~

~~Feature~~	~~How It Works~~
~~Lifestyle quiz~~	~~Sleep schedule, cleanliness, noise, cooking, guests, budget~~
~~AI compatibility matching~~	~~Algorithm + LLM generates compatibility narrative~~
~~Verified profiles~~	~~University email + simple face check~~
~~Joint flat search~~	~~Matched pairs search listings together~~
~~AI chatbot~~	~~"What area should we live in?" → neighbourhood recommendations~~

~~Pros: Solves the compatibility problem first. Most natural user flow. Cons: Requires critical mass of users to find matches.~~

~~Solution B: "TrustFirst" — Verification-First Approach~~
~~Core idea: Every listing and every person is verified before you commit.~~
~~Target: trust between users~~

~~Feature~~	~~How It Works~~
~~AI listing analysis~~	~~Photo/video → AI estimates real size, flags scams, compares price~~
~~3D room reconstruction~~	~~Video walkthrough → rough 3D model + dimensions~~
~~User verification~~	~~University email + face scan~~
~~Landlord verification~~	~~ID check + property ownership confirmation~~
~~Review system~~	~~Tenants review flats and landlords after staying~~

~~Pros: Directly addresses the trust problem. Strongest differentiator. Cons: 3D reconstruction is technically complex. Verification requires data.~~

Solution C: "AllInOne" — Combined Platform
Core idea: Housing search + roommate matching + trust signals in one platform.
→ local only info (which areas more dangerous) + chatbot + online forum
Target: both

Feature	How It Works
AI Housing Chatbot	"I need a room near HKU, under $5k, quiet" → results + advice
Roommate Matching	Lifestyle quiz → compatibility score + narrative
Listing Scanner	Photo → AI extracts info, flags red flags, compares price
Room Size Estimation	Video/photo → AI estimates real dimensions with confidence scores
Virtual Visit	3D model or photo-based room analysis for remote students
Verified Community	University email verification + reviews

Pros: Solves all problems at once. Most comprehensive. Best pitch story. Cons: Most complex to build. Risk of doing nothing well.

User flow:

1. Register
    a. Email (university)
    b. Basic info: name, age, nationality, university
    c. Set password
2. Verify student identity
    a. Upload student card
    b. University email verification
    c. Mark as verified / pending / declined
3. Complete quiz
    a. Lifestyle (roommate compatibility)
    b. Housing preferences (listing matching)
    c. Budget and logistics
    d. UI display results: lifestyle profile (lifestyle, budget, preferred area, room size)
4. AI processing
    a. Lifestyle profile
    b. Compatibility scores with other users

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For each pair of users (A, B):
  1. Compare quiz answers (weighted)
  2. Calculate similarity score (0-100%)
  3. Apply budget filter: |A.budget - B.budget| < 20%
  4. Apply area filter: at least 1 overlapping district
  5. Generate AI narrative explaining the match
  Compatibility Score =
    (A1_match × 3) + (A2_match × 3) + (A3_match × 2) +
    (A4_match × 2) + (A5_match × 1) + (A7_match × 2) +
    (A8_match × 1) + (C2_match × 3) / max_possible × 100
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    c. Filter listings matching preferences
    d. Generative AI narrative for matches
5. Dashboard
    a. UI
    b. AI Chatbot
6. Connect and act
    a. Message matched roommates
    b. View listing details
    c. Virtual visit

***contract signing info shall be provided, e.g. scams and stuff

Can meet up if they want via chat

AI Components

Component	What It Does	Model
Student Card OCR	Extract name, uni, ID from photo	Vision model (GPT-4o-mini or similar)
Roommate Compatibility Narrative	Generate human-readable match explanation	LLM (WorkBuddy/GPT-4o-mini)
Listing Analysis	Extract info from listing photo/screenshot	Vision + LLM
Room Size Estimation	Estimate real dimensions from photos	Vision model + geometry
Price Fairness Check	Compare listing price to market data	LLM + data lookup
Neighbourhood Guide	Describe any district: safety, commute, food	RAG + LLM
Housing Chatbot	Answer questions about housing in any language	LLM (multilingual)
Scam Detection	Flag suspicious listings	LLM + rules

What You're NOT Building (Scope Control)

Feature	Why Not	Alternative
3D room reconstruction	Too complex, 3-4 weeks alone	Photo analysis + AI description
Face scanning	Privacy concerns, complex	University email verification only
Real-time chat	Needs moderation, reporting	Use Telegram/WhatsApp, link out
Payment system	Out of scope for hackathon	Users pay landlord directly
Landlord portal	Extra user type, too much scope	Landlords list via form
Mobile app	Not enough time	Responsive web app
Multi-language UI	Focus on English first	Chatbot answers in any language

Lifestyle Quiz

Section A: Lifestyle (10 questions)

These determine roommate compatibility:

#	Question	Options	Weight
A1	What's your sleep schedule?	Early bird (sleep by 10pm) / Night owl (sleep after midnight) / Flexible	High
A2	How clean do you keep your space?	Very tidy (everything in place) / Moderate (mostly clean) / Relaxed (clutter is fine)	High
A3	How often do you have guests over?	Rarely / 1-2 times a week / Several times a week / Daily	High
A4	How do you feel about noise at home?	Need silence / Background noise is fine / Music/conversation is welcome	Medium
A5	Do you cook at home?	Every meal / A few times a week / Rarely / Never (eat out)	Medium
A6	How do you feel about sharing food?	Happy to share / okay sometimes / Prefer separate	Low
A7	What's your approach to shared spaces?	Clean immediately after use / Clean daily / Clean weekly / Whoever made the mess handles it	High
A8	How do you handle conflicts?	Discuss directly / Avoid confrontation / Need time to cool off first	Medium

A9	Are you okay with pets?	Love them / Okay with some / Prefer no pets / Allergic	Low
A10	What's your work/study from home situation?	Often home during day / Usually out / Hybrid / Varies	Medium

Section B: Housing Preferences (8 questions)

These determine listing matching:

#	Question	Options	Weight
B1	Which districts do you prefer?	Multi-select: Wan Chai, Central, Sai Ying Pun, Causeway Bay, Tsim Sha Tsui, Mong Kok, Sham Shui Po, Prince Edward, Kowloon City, Wong Tai Sin, Diamond Hill, Choi Hung, Sau Mau Ping, Tseung Kwan O, Tai Po, Sha Tin, Ma On Shan, Tsuen Wan, Yuen Long, Tuen Mun, Kennedy Town, Pok Fu Lam	High
B2	What room type do you want?	Private room in shared flat / Master ensuite / Whole studio / Whole flat	High
B3	What's your minimum room size?	Any / >60 sq ft / >80 sq ft / >100 sq ft / >120 sq ft	Medium
B4	How important is MTR proximity?	Must be within 5 min walk / Within 10 min / Within 15 min / Don't mind	Medium
B5	What floor do you prefer?	Lower (1-5) / Mid (6-15) / High (15+) / No preference	Low

B6	Do you need specific amenities?	Multi-select: Air conditioning, Washing machine, Elevator, Balcony, Kitchen, Furniture, WiFi	Medium
B7	When do you need to move in?	Immediately / Within 1 month / Within 2-3 months / Flexible	High
B8	How long is your lease?	6 months / 1 year / 1-2 years / Flexible / Not sure yet	High

Section C: Budget & Logistics (4 questions)

These determine affordability matching:

#	Question	Options	Weight
C1	What's your monthly budget for rent?	Slider: $3,000 - $15,000 HKD	Critical
C2	Are you open to split rent with a roommate?	Yes, looking to share / Prefer my own place / Flexible	Critical
C3	Do you have a guarantor in HK?	Yes / No / Not sure	Low
C4	Are you an exchange student?	Yes (duration: ___) / No, full-time student / No, working professional / Other	Medium